THE LOCUS WHERE ALL TWISTS BY DN ARE INVERTIBLE WITH ISOMORPHIC MULTIPLICATIONS

THE STACKS PROJECT AUTHORS

Editorial note. Both the chart-inclusion step and the final multiplication step say 'by arguing as in / compare with the proof of Lemma when-invertible'; the when-invertible card carries the full template, and the new content (the converse chart construction) is displayed in-span.

Allowed standard prerequisites (editorial).

- Trivialization on standard opens. lemma-when-invertible (carded this batch). Template for arguing invertibility from generating sections; cited twice in the proof.

Source reference. Original equation 01MO: sources/stacks-constructions/pinned-original-source.tex, source lines 1930–1930. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original equation 01MQ: sources/stacks-constructions/pinned-original-source.tex, source lines 1966–1966. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 01MS: sources/stacks-constructions/pinned-original-source.tex, source lines 1986–1986. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a graded ring. Set $X = Proj(S)$. Fix $d \geq 1$ an integer. The following open subsets of X are equal:*

1. *The largest open subset $W = W_d \subset X$ such that each $\mathcal{O}_X(dn)|_W$ is invertible and all the multiplication maps $\mathcal{O}_X(nd)|_W \otimes_{\mathcal{O}_W} \mathcal{O}_X(md)|_W \to \mathcal{O}_X(nd+md)|_W$ (see 01MO) are isomorphisms.*
2. *The union of the open subsets $D_+(fg)$ with $f, g \in S$ homogeneous and $\deg(f) = \deg(g) + d$.*

Moreover, all the maps $\widetilde{M}(nd)|_W = \widetilde{M}|_W \otimes_{\mathcal{O}_W} \mathcal{O}_X(nd)|_W \to \widetilde{M(nd)}|_W$ (see 01MQ) are isomorphisms.

Date: stacks-project:constructions.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. If $x \in D_+(fg)$ with $\deg(f) = \deg(g)+d$ then on $D_+(fg)$ the sheaves $\mathcal{O}_X(dn)$ are generated by the element $(f/g)^n = f^{2n}/(fg)^n$. This implies x is in the open subset W defined in (1) by arguing as in the proof of Lemma 01MS.

Conversely, suppose that $\mathcal{O}_X(dn)$ is free of rank 1 for all n in an open neighbourhood V of $x \in X$ and all the multiplication maps $\mathcal{O}_X(nd)|_V \otimes_{\mathcal{O}_V} \mathcal{O}_X(md)|_V \to \mathcal{O}_X(nd+md)|_V$ are isomorphisms. We may choose $h \in S_+$ homogeneous such that $x \in D_+(h) \subset V$. By the definition of the twists of the structure sheaf we conclude there exists an element s of $(S_h)_d$ such that s^n is a basis of $(S_h)_{nd}$ as a module over $S_{(h)}$ for all $n \in \mathbf{Z}$. We may write $s = f/h^m$ for some $m \geq 1$ and $f \in S_{d+m\deg(h)}$. Set $g = h^m$ so $s = f/g$. Note that $x \in D_+(g)$ by construction. Note that $g^d \in (S_h)_{d\deg(g)}$. By assumption we can write this as a multiple of $s^{\deg(g)} = f^{\deg(g)}/g^{\deg(g)}$, say $g^d = a/g^e \cdot f^{\deg(g)}/g^{\deg(g)}$. Then we conclude that $g^{d+e+\deg(g)} = af^{\deg(g)}$ and hence also $x \in D_+(f)$. So x is an element of the set defined in (2).

The existence of the generating section $s = f/g$ over the affine open $D_+(fg)$ whose powers freely generate the sheaves of modules $\mathcal{O}_X(nd)$ easily implies that the multiplication maps $\widetilde{M}(nd)|_W = \widetilde{M}|_W \otimes_{\mathcal{O}_W} \mathcal{O}_X(nd)|_W \to \widetilde{M(nd)}|_W$ (see 01MQ) are isomorphisms. Compare with the proof of Lemma 01MS. $\square$